GEK-88203

DRIVE SYSTEM INSTRUCTIONS

CONRAC MONITOR

INSTALLATION & MAINTENANCE MANUAL

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

CONRAC MONITOR

INSTALLATION AND MAINTENANCE MANUAL

(GEK-88203)

GEK-88203

ISSUED: September 1983

NOTICE

THIS MANUAL IS A REPRODUCTION OF SELECTED PORTIONS OF THE "CONRAC INSTALLATION, OPERATION, AND MAINTENANCE MANUAL" (SNA SERIES VIDEO MONITORS) WHICH ARE BELIEVED TO SERVE THE PRIMARY NEEDS OF THE USER. COMPLETE COPIES OF THE CONRAC MANUAL CAN BE OBTAINED FROM:

CONRAC DIVISION
CONRAC CORPORATION
600 N. RIMSDALE AVENUE
COVINA, CA 91722

NOTICE

The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Department, 1501 Roanoke Boulevard, Salem, Virginia, USA, 24153.

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*(SNA-23 uses the same chassis)

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SECTION 1

SPECIFICATIONS

VISUAL PERFORMANCE

Linearity and Geometry	No point on raster deviates from its proper position by more than 2% of raster height.
Brightness and Resolution	Minimum TV lines 800 center, 700 corner at 50 fL luminance.
Phosphor Colorimetry	P4 phosphor, 9300°K (others available on special order).
Interlace	Better than 90%.
Raster Size Regulation	Less than 1% change, 0% to 100% APL (Average Picture Level) at 50fL peak luminance.
Black Level Stability	Switchable dc restorer maintains black level shift less than 2% of peak luminance from 10% to 90% APL.
Discernible Shades of Gray	10, minimum.
Dynamic Focus	Standard on SNA17 and SNA23 only.

OPERATING ENVIRONMENT

Ambient Temperature	10°C to 50°C operating
Humidity	10% to 90%, non-condensing

INPUT POWER REQUIREMENTS

Voltage	Depends upon power transformer connections. Refer to Section 2.
Frequency	50-60 Hz.
Power	90 watts @ 120 volts, 50/60 Hz.
Connection	Captive 6 foot cord with 3-prong connector plug.

SIGNAL INPUTS

Video	Composite: loop-through or switchable to internal 75 ohm termination, 1.0V p-p nominal (.35V to 2V), sync negative.
Video	Non-Composite: 0.7V p-p nominal (.25V to 1.4V) black negative.
Return Loss	Greater than 40dB.
External Sync	4V p-p nominal (1V to 8V) negative.

CIRCUIT PERFORMANCE

Video Amplifier Frequency Response	± 1 dB, 60 Hz to 10 MHz (100 kHz reference). Lessthan 6 dB down at 15MHz.
Video Amplifier Differential Gain	Less than 5% for luminance range of 0 fL to 50 fL
Vertical Retrace Time	1000 microsec, maximum.
Horizontal Retrace Time	10 microsec, maximum.
Low Voltage Regulation	Less than 1% change for variations from nominal line voltage of ± 10%.
High Voltage Stabilization	To maintain less than 2% change for 0% to 100% APL at 50 fL peak luminance.

MECHANICAL

Weights, Net and Shipping

Model	Net Weight	Shipping Weight
SNA14/R, 15/R	20.5 kilos (45 lbs.)	26.9 kilos (59 lbs.)
SNA23Y	41.2 kilos (91 lbs.)	49.2 kilos (108 lbs.)

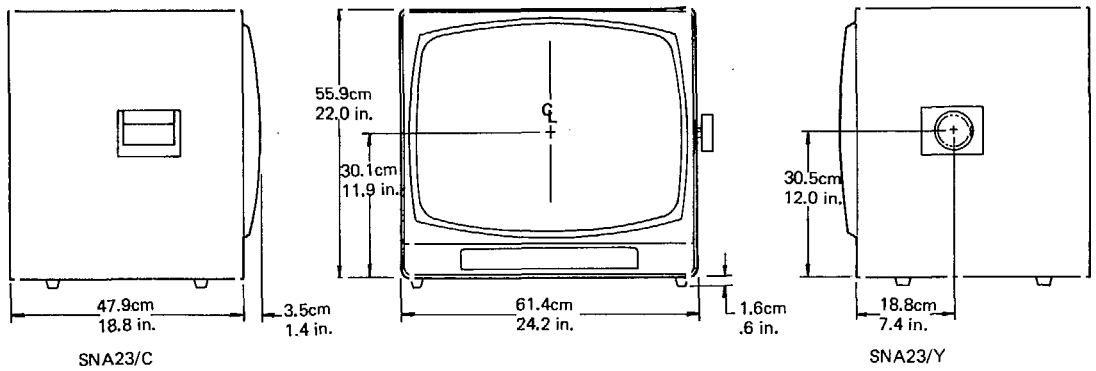


Figure 1-1. Outline Dimensions, Cabinet Models

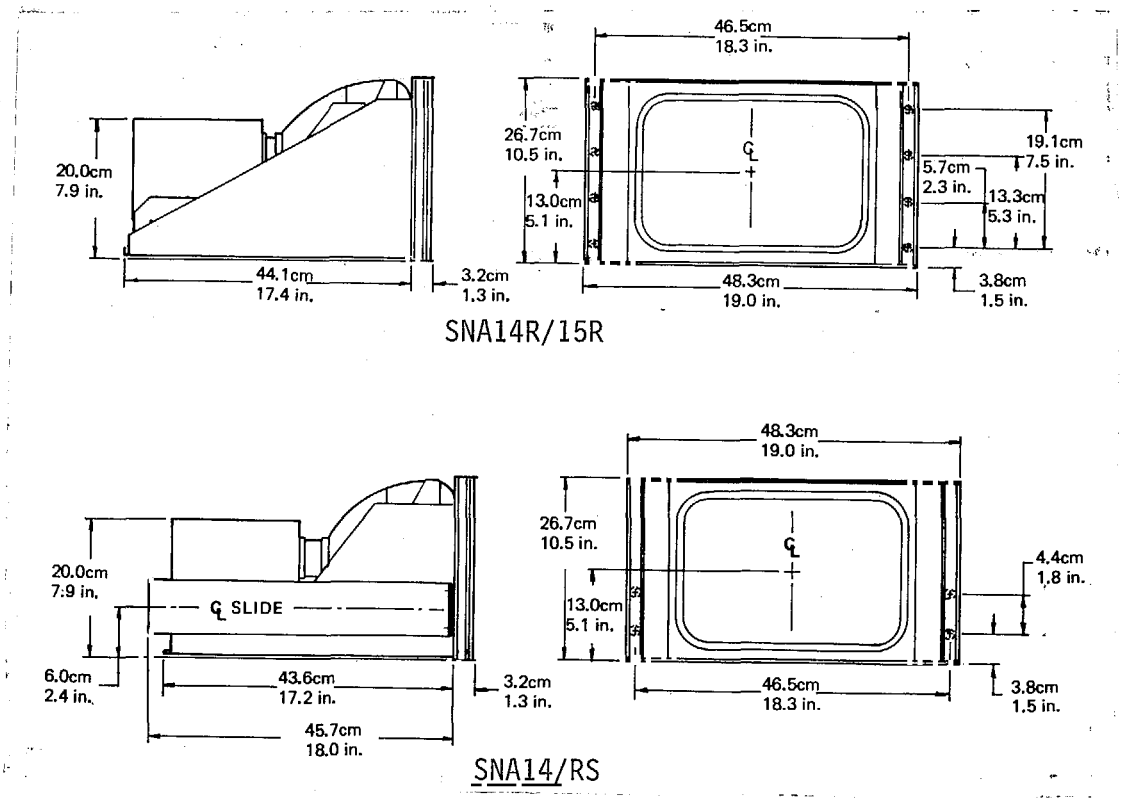


Figure 1-2. Outline Dimensions, Rack-Mount Models

NOTES

SECTION 2

INSTALLATION AND OPERATING INSTRUCTIONS

Unpacking

Carefully remove all packing material and inspect the equipment for any damage which may have occurred during shipment. Report any shortage or damage to the carrier.

Installation, Physical

Install your SNA monitor in a clean location, free from moisture and excessive heat. Be sure to provide adequate ventilation so the unit will not overheat because of its own internal heat generation during operation. During installation use caution to avoid striking the picture tube. If your unit is not supplied in a cabinet, be particularly careful of the picture tube neck.

Installation, Electrical

****
* WARNING *
****

THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THESE INSTRUCTIONS SHOULD INSTALL, OPERATE, OR MAINTAIN THIS EQUIPMENT.

****
* WARNING *
****

TO PREVENT HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MUST BE ADHERED TO.

****
* CAUTION *
****

BEFORE CONNECTING YOUR MONITOR TO THE ELECTRICAL POWER LINES (MAINS), BE CERTAIN THAT IT IS CORRECTLY WIRED FOR THE VOLTAGE SUPPLIED IN YOUR AREA.

All SNA series monitors (built after August, 1977) are supplied with a power transformer (PN 965112) which has two tapped primary windings. This transformer can be wired, depending upon how the primary is connected, for almost any standard power line (mains) voltage found anywhere in the world. All standard order SNA series monitors are supplied with the transformer primary connections wired for 120 volt operation. SNA series monitors built for use on power line voltages other than 120 V are handled as special order items and will have the primary connections of the transformer wired as stipulated in the special order. Figure 2-1 illustrates the proper primary connection for the transformer for any of the world's common line voltages.

SNA series monitors built before August, 1977, were supplied with one of two different power transformers. Standard order units were supplied with transformer No. 965070, which has two non-tapped primary windings. This transformer was used in units built for 120 volt service (with the two primary windings tied in parallel across the ac power line), or for 240 volt service (with the windings series connected). Units built before August, 1977, which were ordered for operation on voltage other that 120 or 240, were equipped with transformer number 965098. This transformer has two tapped primaries and can be wired for any standard line voltage. Figure 2-4 illustrates the proper primary connection for both the 965070 and 965098 transformers for any voltage on which they will operate.

After determining that the power transformer and its primary wiring are correct for the line voltage available in your area, check the fuse and verify that it is of the correct rating for the line voltage. Refer to Table 2-1 for fuse information.

Table 2-1. Fuse Data

Model	100-120V	200-240V
SNA-14R/15R	1A	0.5A
SNA-23	1.5A	0.75A

Input Signal Connections

All input signals are made at the rear of the chassis. In cabinet-mounted units, the rear panel does not cover the rear of the chassis. This allows access to the chassis rear. Figure 2-2 illustrates both the front and rear panels of the various standard configurations of the SNA series monitors. Refer to the correct part of Figure 2-2 which depicts the rear panel of your particular unit. If yours is a non-standard, or special order unit, it may differ in appearance somnewhat from the drawing shown in Figure 2-2.

Input Signal Connections (Continued)

In all SNA series monitors except some special order units equipped with dual inputs, the two video connectors (9) are loop-through wired BNC connectors. One is labeled IN and the other is labeled OUT. Associated with the two video connectors in back of each unit is the input impedance selection switch (10). The switch has two positions, HI and 75 ohms. With this switch in the HI position, the input impedance of the monitor is high. In the 75 ohm position, the input impedance is terminated in 75 ohms. It is important that the end of any video transmission line be terminated in its characteristic impedance to eliminate reflections which would cause ringing and other picture degradations. Usually, video signals are transmitted along 75 ohm coaxial lines. If this is the case and the SNA monitor is the end item in the line, it can be easily terminated by placing the switch in the 75 ohm position and supplying the input signal to the VIDEO IN connector. In a loop-through configuration, where other equipment is connected to the same line and the SNA is not at the end of the line, the switch should be placed in the HI position. The signal is supplied to the VIDEO IN connector and taken from the OUT connector to be supplied to the next unit in the loop-through chain. Whatever the case, termination must be made at the end of the line.

Sync, Internal/External

The SNA series monitors are designed to operate from either composite video (a video signal containing sync signals) or from non-composite video (a video signal which contains no sync signals). When the unit is operating from a non-composite video signal source, the sync signals must also be separately supplied to the monitor. On the back of each SNA monitor are two sync input connectors (11). These are parallel-connected BNC connectors and are associated with the sync selector switch (12). For composite video operation, the sync selector switch (12) is placed in the INT position and no connection to either of the sync inputs (11) is necessary. For non-composite video operation, the sync selector switch is placed in the EXT position and mixed sync must be supplied to either one or both of the two sync input connectors (11). The sync inputs do not have to be terminated in 75 ohms except in certain cases, so no impedance selection switch is associated with them. If, in a given situation, it is deemed necessary to terminate a sync input line, the input can be supplied to one connector, and a termination plug can be connected to the other.

Operating Controls

All operating controls on SNA series monitors, with the exception of the dynamic focus amplitude control, are found on the front of the monitor. On the 14-inch rack mount model, they are located on each side of the picture tube. On all other units, they are located below the picture tube. Figure

Operating Controls (Continued)

2-2 illustrates both the front and rear panels to the various standard configurations of the SNA series monitors. Refer to the correct part of Figure 2-2 which depicts the front panel of your particular unit. If yours is a non-standard, or special order unit, it may differ in appearance somewhat from the drawing shown in Figure 2-2. The function of each of the operating controls is described below:

(1) Power ON/OFF switch ... switches the ac power applied to the unit

(2) Brightness control ... adjusts the brightness level of the CRT raster

(3) Contrast control ... adjusts the contrast level of the picture

(4) Vertical hold control ... adjusts the free-running frequency of the vertical oscillator

(5) Height control ... adjusts the vertical size of the CRT raster

(6) Focus control ... adjusts the sharpness of the raster scan lines on the CRT

(7) Horizontal hold control ... adjusts the free-running frequency of the horizontal oscillator

(8) Width control ... adjusts the width of the CRT raster

(13) Dynamic focus amplitude control ... This control, found only on the 17- and 23- inch units, is used in conjunction with the focus control (6) to adjust the sharpness of the CRT raster scan lines at the edges of the screen.

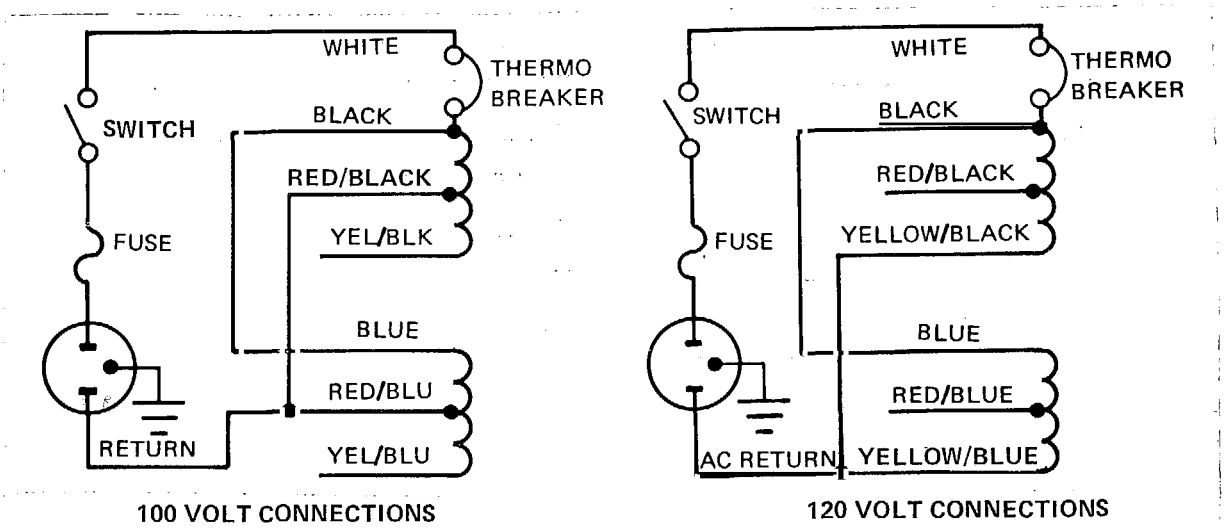


Figure 2-1A. Primary Connections for Type 965122 Power Transformer

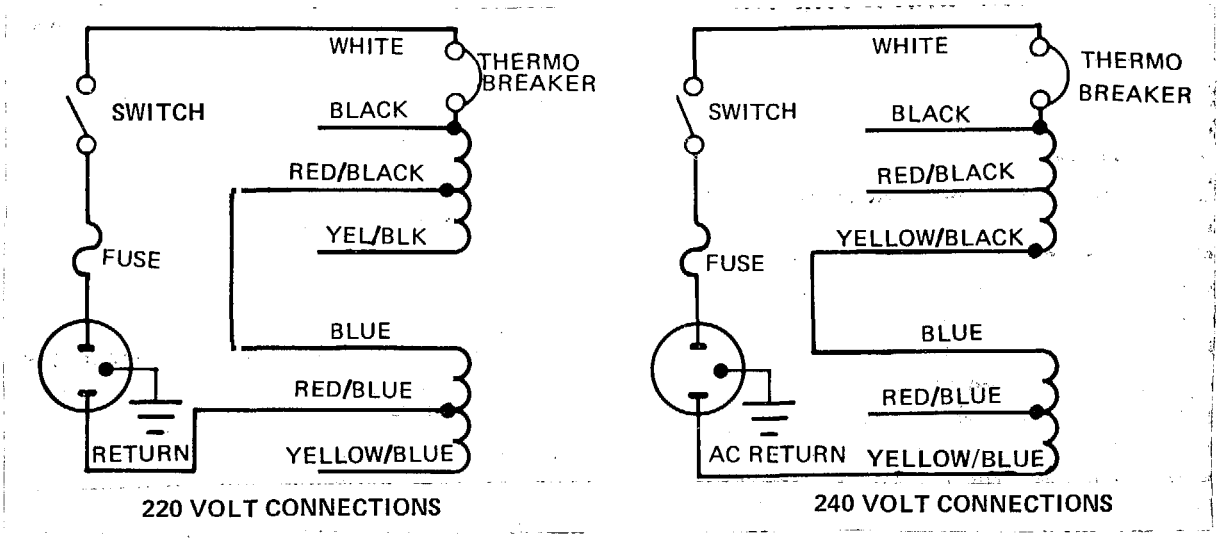


Figure 2-1B. Primary Connections for Type 965122 Power Transformer

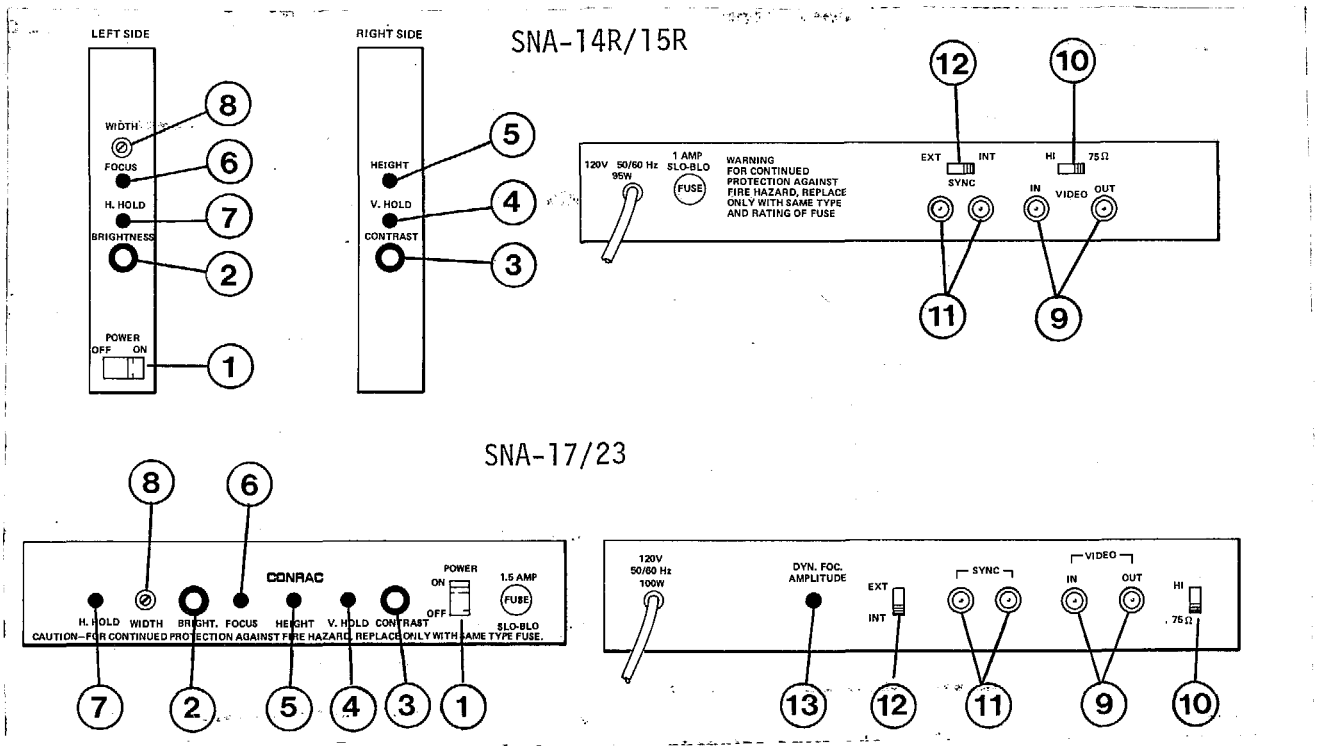


Figure 2-2. SNA Front and Rear Panels

Operating Instructions

Install and connect the units as described above. Turn on power switch (1).

Operating Instructions (Continued)

After the monitor has warmed up enough to display a picture, adjust, if necessary, the horizontal hold control (7) and the vertical hold control (4) so that the displayed picture is stable. Set the brightness control (2) and the contrast control (3) to display a pleasant picture of fairly low brightness level. Then allow the monitor to warm up for a minimum of 15 minutes before making final adjustments.

After the monitor is warmed up, adjust the brightness (2) and contrast (3) controls for the desired level of brightness and contrast. Remember, picture tube life is proportional to brightness level. Don't use more brightness than is needed. (This precaution is particularly true for special order units supplied with special CRT phosphors. In some cases, operating a tube at high brightness level can damage the screen within a few hours.) Adjust the height (5) and width (8) controls for the desired height and width. For underscan operation, refer to "Underscan Operation" in this section of the manual. Adjust the focus control (6) for optimum focus of the raster scan lines at the center of the screen. Then (on 17- and 23-inch units, only) adjust the dynamic focus amplitude control (13) for optimum focus of the raster scan lines at the center edges of the screen.

To adjust the vertical hold, rotate the vertical hold control (4) so that the picture is out of sync and rolling up (not down). Rotate the control to the point where the picture just locks in sync, then a fraction of a turn past that point.

To adjust the horizontal hold, it is necessary to first disable the sync. If your monitor is operating from composite video, insert a 75 ohm termination plug into one of the two sync input connectors (11) and place the sync selector switch (12) in the EXT position. If your monitor is operating from non-composite video, simply disconnect the external sync from the sync input and insert a 75 ohm termination plug in its place. (The 75 ohm termination plug is needed to prevent the possibility of the horizontal circuit trying to lock up on its own flyback pulse.) With the sync disabled, adjust the horizontal hold control (7) so that the horizontal oscillator is almost exactly on frequency. The horizontal blanking bar should be straight and vertical, floating slowly from side to side. After the control has been so adjusted, restore the sync. The picture should immediately snap into sync and remain locked.

Underscan Operation

All standard SNA series monitors are designed to be capable of displaying a raster which is overscanned (typically 3-5%) or underscanned (typically 10%). In changing from overscan to underscan, or vice-versa, the vertical aspect of the raster is entirely adjusted by means of the height control (5). The height control has no effect upon the vertical linearity. The horizontal aspect of the picture size is adjusted in all standard late model SNA series

Underscan Operation (Continued)

monitors by means of a five-position plug and jack arrangement located on the circuit board inside the high voltage module. To select overscan, underscan or square raster operation, turn off the unit and open the high voltage module by removing the retaining screw and sliding up the top/side panel. Select the correct position for jack J-7 according to Table 2-2.

SNA series monitors that have been supplied (on special order) for high line rate operation are equipped with the type 106368 high voltage module. In this high voltage module, jack J-7 has only three positions. These are labeled FULL, UNDER and SQUARE on the circuit board. The wiring from the flyback transformer to jack J-7 varies according to the line rate and picture tube size. This has been set at the factory for the CRT size and line rate specified in the special order against which the monitor was originally built.

Table 2-2. J-7 Connections

Unit Type	Type of Operation	Connect J-7 to
SNA-14R/15R	Overscan	B
	Underscan	C
SNA-23 (90°)	Overscan	A
	Underscan	B
	Square Raster	C
SNA-23 (110°)	Overscan	A (use width coil)
	Underscan	A (use width coil)
	Square Raster	B

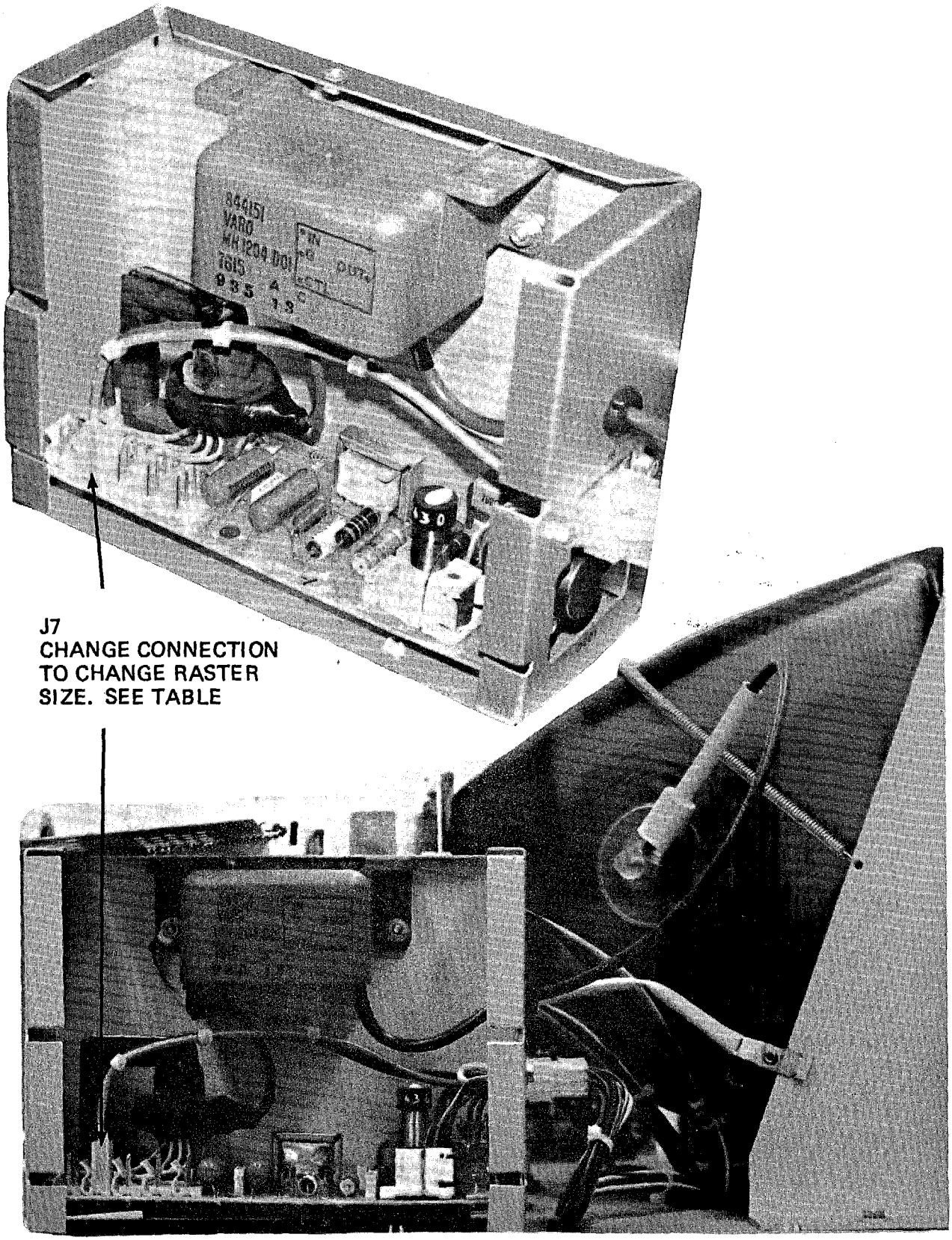
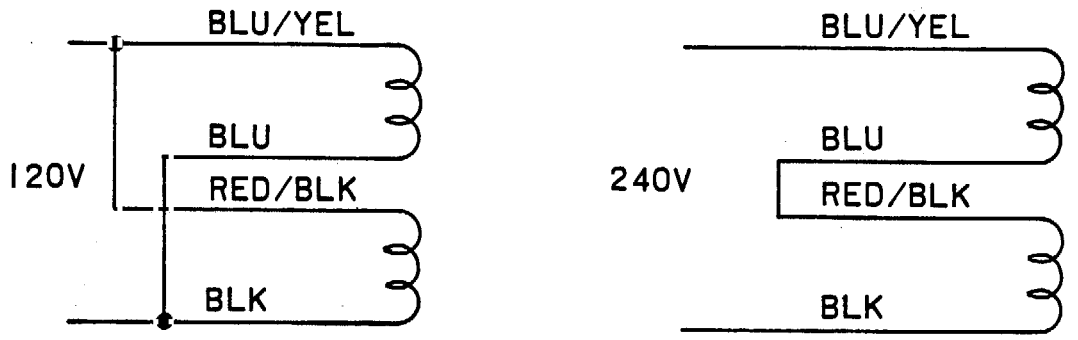
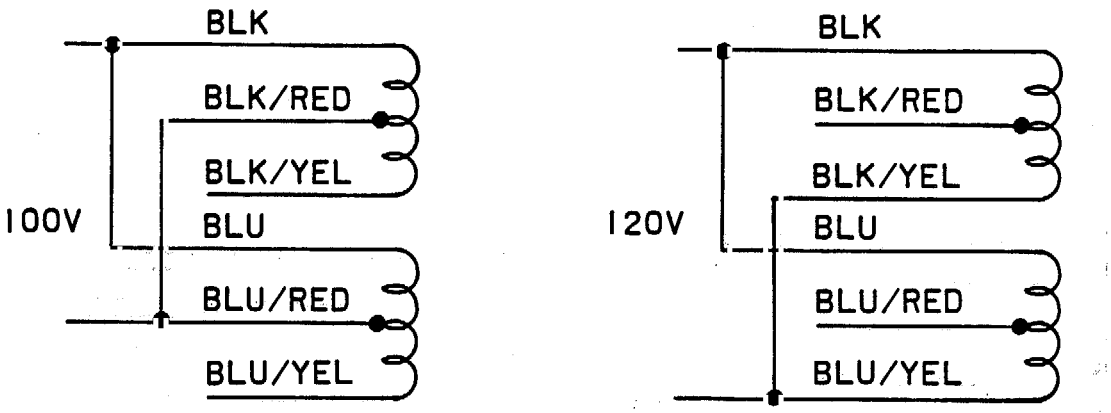
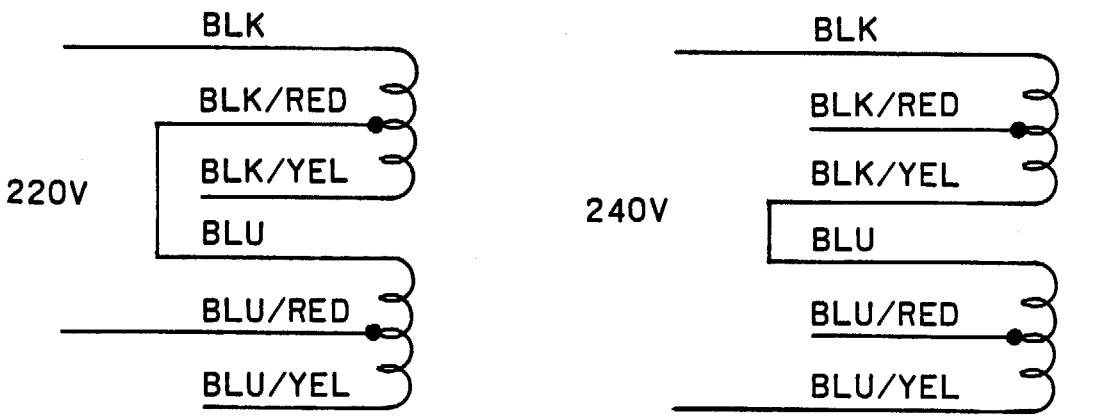


Figure 2-3. Location of Jack J7 in High Voltage Module



Primary connections for transformer 965070





Primary connections for transformer 965098

Figure 2-4. Primary Connections for Transformers Used Before October, 1977

NOTES

SECTION 3

MAINTENANCE AND ALIGNMENT INSTRUCTIONS

Replacement Parts

There is no single parts list for the entire SNA Monitor series. These monitors consist of a series of subassemblies, or modules, each of which has its own parts list.

For replacement parts for a particular series, contact:

CONRAC CORPORATION
600 N. Rimsdale Avenue
Covina, CA 91722

****
* WARNING *
****

ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENT'S CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

Disassembly and Maintenance

In general, the SNA series monitors do not require routine preventive maintenance. When repair is needed, and the unit must be removed from the cabinet, proceed as follows:

1. Unscrew the retaining screws which hold the back on the cabinet and remove the back. (Late model, large screen units use twist-lock quick release devices instead of screws.)

2. Remove any knobs from shafts which protrude through the cabinet front.

3. Tilt the cabinet up (from the back) and remove the bolts which secure the chassis to the cabinet.

Disassembly and Maintenance (Continued)

4. Slide the chassis out the back of the cabinet. On all except the 23-inch version, the CRT is attached to, and comes out with, the chassis.

All modules and circuit boards can be easily removed from the chassis when necessary for component replacement. They are secured to the chassis by screws, and all electrical connections are make through push-on connectors.

Picture Tube Replacement

In all versions of the SNA except the 23-inch model, the picture tube is mounted to the chassis. In the 23-inch model, the picture tube is made with four mounting ears, one at each corner, which are permanently attached to the tube. The tubes are bolted to the cabinet with four bolts which go through holes in these mounting ears. To replace the picture tube in the 23-inch model, proceed as follows:

1. Remove the chassis from the unit.

2. Remove the deflection yoke and centering ring assembly from the neck of the picture tube.

3. Place the cabinet face down on a padded surface.

4. Remove the four mounting bolts at each corner of the picture tube.

5. Reach into the cabinet and lift out the old tube by the bell. Do not lift the tube out by the neck.

6. Install the replacement tube by reversing the procedure of the above 5 steps. Be sure to orient the ultor terminal in the same position as the original tube.

In all versions of the SNA where the picture tube is mounted to the chassis, it is secured to two vertical brackets on the chassis by four mounting screws, one in each corner. In earlier units, the picture tube is mounted by four mounting ears, one at each corner, which are held in the four corners of the tube by a tension band which goes around the front bell of the tube. In later production units, the mounting method was changed to a "clamshell" configuration in which the bell of the tube is sandwiched between two halves of the clamshell. One half goes over the bell from the neck end and the other goes over the bell from the front. In each corner, there are holes for the four mounting screws. These holes go through both halves of the clamshell into the two vertical members of the chassis.

Picture Tube Replacement (Continued)

To replace the picture tube in early models which do not have the clamshell, proceed as follows:

1. Remove the chassis from the unit.

2. Loosen the screw in the clamp that holds the deflection yoke assembly on the neck of the tube. Some units also have a resoultion magnet assembly mounted on the tube neck near the socket. If your unit is so equipped, loosen this clamp also.

3. Remove the CRT socket and high voltage connector from the tube.

4. Remove the four mounting screws that secure the tube to the two vertical members of the chassis. Hold the tube in place by the neck when removing the four screws to prevent any sudden tipping.

5. Gently slide the tube forward away from the chasis, slipping the neck components off the neck as the tube is brought forward. Avoid touching the ultor terminal. It may still contain residual high voltage and the sudden shock, though usually harmless, may cause you to drop the tube.

6. Place the tube face down on a padded surface and loosen the mounting band which holds the four mounting ears onto the tube. Remove the mounting band and ears from the old tube.

7. To install the new tube, reverse the procedures of the above steps. Be sure that the mounting ears align with the mounting holes before tightening the band around the tube. Be sure to orient the ultor terminal in the same position as the original tube.

To replace the picture tube in later models which have the clamshell, proceed as follows:

1. Perform steps 1 through 4 above, in order, for tubes which do not have the clamshell.

2. Gently slide the tube forward, together with both halves of the clamshell, and pull it away from the chassis. Slip the neck components off the tube as it is brought forward. Avoid touching the ultor terminal. It may still contain residual high voltage and the sudden shock, though usually harmless, could cause you to drop the tube.

3. Remove both halves of the clamshell from the front and back of the old tube.

4. To install the new tube, reverse the procedures of the above steps. Be sure to orient the ultor terminal in the same position as the original tube.

Picture Tube Replacement (Continued)

After replacement of any picture tube (Refer to Figure 3-1)

- Adjust the position and angle of the deflection yoke on the tube neck so that the picture is straight, then tighten the yoke clamp.
- Now adjust the centering rings on the yoke assembly to center the raster both vertically and horizontally.

If the unit is equipped with a resolution magnet assembly

1. set both tabs of the resolution magnet assembly together and up, and
2. perform the focus and dynamic focus adjustments (outlined in a subsequent part of this section of this manual).

After focus adjustments are made, turn the front panel focus control all the way to one end to throw the center screen out of focus, then spread and turn the tabs, as necessary, to obtain the best apparent corner focus in the four corners.

- Reset the center screen focus.

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* * * * * *
* CAUTION *
* * * * * *

* * * * * * * * * * * * * * * * * * * * * * * * *
* ALL CONRAC MODEL SNA MONITORS USE SPECIAL     *
* PICTURE TUBES AVAILABLE ONLY FROM CONRAC.     *
* DO NOT ATTEMPT TO USE TUBES INTENDED FOR THE  *
* DEALER SERVICE REPLACEMENT MARKET.            *
* * * * * * * * * * * * * * * * * * * * * * * * *
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Table 3-1. Required Test Equipment

REQUIRED FOR	DESCRIPTION OF REQUIRED ITEM	
Measurement and adjustment of the power supply voltage	3 1/2 digit digital multimeter	The basic requirement is that the meter measure accurately, within 1/2% of reading. The Fluke 8000 A is a low-cost unit which is satisfactory.
Adjustment of the vertical and horizontal oscillator's free running frequencies	Source of a stable video signal	This may be from a video camera, a dc modulated broadcast signal, or a video signal generator. The signal source must be capable of delivering 1 volt p-p into 75 ohms. The Visual Information Institute, Model 207 Bar/Dot Mixer is a low-cost unit which satisfies the requirement.
Adjustment of the vertical and horizontal linearity	Crosshatch video signal source	This may be from a video switcher or from a separate signal generator. The signal source must supply composite video at 1 volt p-p into 75 ohms. The Visual Information Institute, Model 207 Bar/Dot Mixer is a low-cost unit which satisfies the requirement.
Alignment of video amplifier bandpass	Video sweep generator	This unit must be capable of supplying 0 to 30 MHz video sweep, with markers at 5 MHz intervals. The signal must be 1 volt p-p into 75 ohms. The Texscan, Model VS-30A is recommended.
Alignment of video amplifier bandpass and adjustment of dynamic focus waveshape	Oscilloscope with 10X and 100X low capacitance probes	The basic requirement is that the vertical passband be totally flat from dc (or close to dc) to 20 MHz. This requires an oscilloscope with a dc to 50 MHz vertical channel. The Tektronix Model 465, or the Phillips Model PM3240X will fulfill this requirement.
Alignment of video amplifier bandpass	Sweep sync compositor	It is necessary to construct your own unit. Follow the schematic diagram in Figure 3-2.

Alignment and Internal Adjustments

Alignment, or adjustment of internal controls, is usually only necessary following repairs to the unit. It is then, usually, only necessary in the particular circuit in which repairs were made. The technician making repairs is expected to use good judgement in determining which area or areas of the unit require any adjustment or alignment.

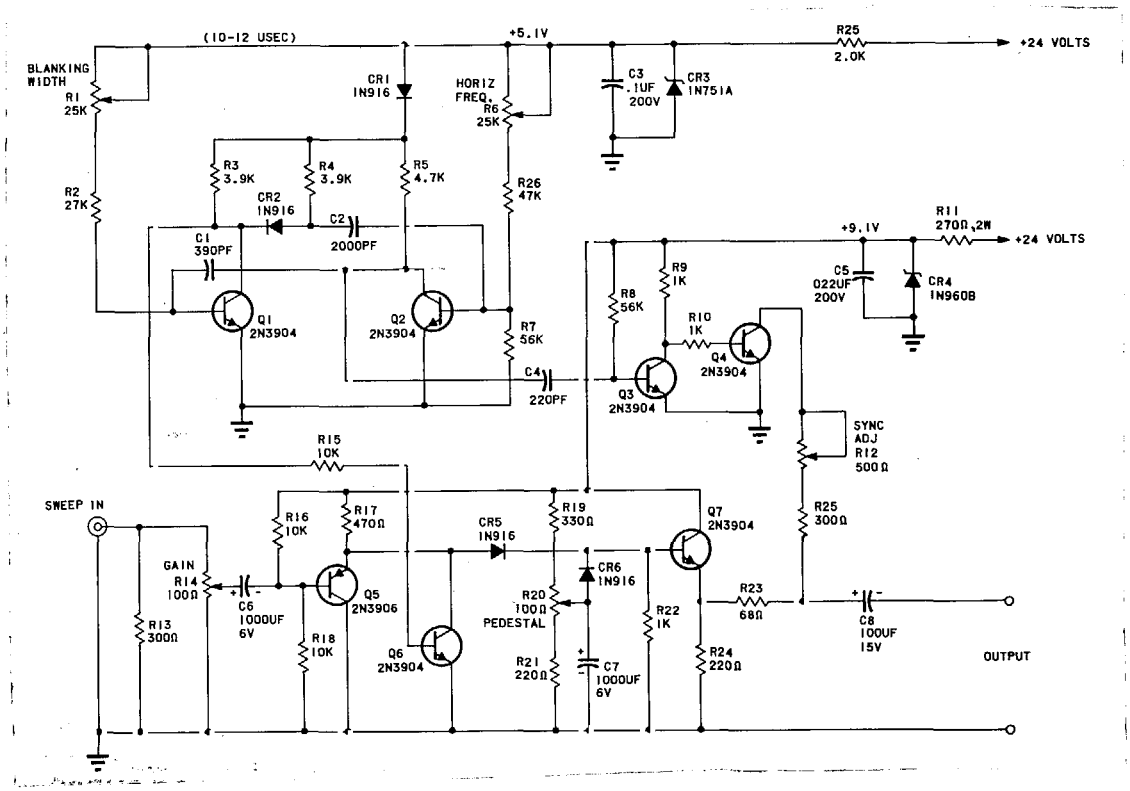


Figure 3-2A. Schematic Diagram of Sync Compositor

Notes:

1. Construct the circuit on vector board, and house the completed unit in a small box chassis. The required +24V power supply (not regulated) may be contained within the same chassis if desired.
2. All resistors and capacitors are ±10% tolerance.
3. Input/output connectors should be BNC or UHF type.
4. Adjust horizontal frequency (R6) to match horizontal line rate of monitor under alignment (15,750 Hz for EIA Standard).
5. Connect sweep input to output of sweep generator and set sweep generator for maximum output. Preset gain control (R14) to minimum.
6. Adjust the remaining controls to obtain the output waveform shown on page 3-8.

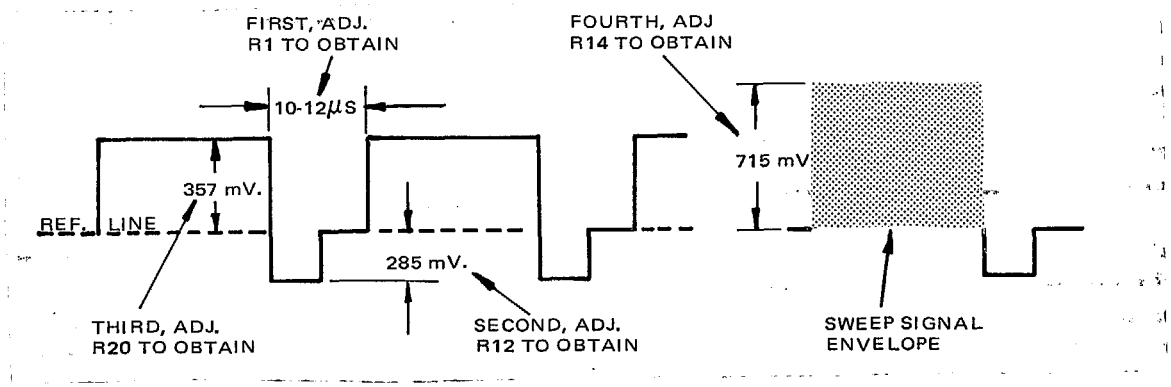


Figure 3-B. Waveform of Sync Compositor

Power Supply Adjustment

1. Plug the monitor into a source of the correct power line (mains) voltage and turn it on.

2. Connect the digital multimeter (set for dc volts) between ground (chassis) and the B+ terminal on the power supply regulator board (Board No. 162400).

3. Adjust pot R10 on the 162400 board to set the voltage to +75 ± 1.5 volts.

   NOTE: On later production units, R10 has been factory sealed to set this voltage. It should be re-adjusted only if a component in the adjustment circuitry or the reference zener has been replaced.

Horizontal Oscillator Frequency Range Adjustment (For standard models operating at a 525 or 625 line rate)

1. With the monitor operating, connect a stable composite video signal source to the video input. Place the 75 ohm termination switch in the 75 ohm position.

2. Pre-set the front panel horizontal hold control to its mid-range position.

3. Disable the horizontal sync by connecting a short clip lead between chassis ground and the junction of resistor R55 and capacitor C28 on the 162393 circuit board. (On older units using the early 162393 circuit board in which the horizontal oscillator circuit is discrete, rather than an integrated circuit, connect the clip lead between the collector and emitter of transistor Q1.)

Horizontal Oscillator Frequency Range Adjustment (Continued)

4. Adjust horizontal frequency control R11 on circuit board 162393 so that the horizontal oscillator is exactly on frequency, and the horizontal blanking bar floats slowly across the screen. The horizontal blanking bar should be vertical and straight.

5. Rotate the front panel horizontal hold control from fully clockwise to fully counter-clockwise and back. Verify that the picture goes out of sync by about the same amount at each extreme, then return the control to the center so that the picture is nearly in sync.

6. Remove the clip lead installed in step 3 above and verify that the picture immediately snaps into sync.

Vertical Oscillator Frequency Range Adjustment

1. With the monitor operating and a stable composite video signal applied, verify that the input is terminated in 75 ohms and the SYNC INT/EXT switch is set to INT. This supplies sync signals to the vertical oscillator.

2. Adjust the front panel vertical hold control so that the picture is out of sync and rolling up (not down).

3. Slowly turn the control to the point where the picture just locks into sync, then just a fraction of a turn further past that point. Leave the control at this setting.

Width Adjustment

The width control is a coil which is screwdriver adjustable, by using an alignment tool, through a hole in the front panel. The width control gives a limited range of adjustment which is generally not sufficient to change the raster from overscan to underscan, or vice versa. The only exception is the SNA-23 with a 110° CRT. In this case, underscan is accomplished by use of the width control. To change between normal and underscan width, it is necessary to change the position of taps on the flyback transformer. Refer to "Underscan Operation" and Table 2-2 in Section 2 of this manual.

Horizontal Linearity Adjustment

1. With the monitor operating, connect a crosshatch composite video signal to the input. Terminate the input in 75 ohms and set the sync selector switch for internal sync operation.

2. Using a tape measure or ruler, measure the horizontal spacing between

Horizontal Linearity Adjustment (Continued)

each adjacent pair of vertical lines in the pattern (the width of each square). Write down each of these measurements.

3. Add up each of the numbers recorded in step 2 and divide the total by the number of measurements taken. This gives the average, or nominal spacing between the vertical lines.

4. Adjust the horizontal linearity coil (located in the high voltage module) to set the lenearity so that the width of each square in the crosshatch pattern is within ± 20% of nominal.

NOTE: The horizontal linearity coil must be adjusted by using a square-shaped non-metallic alignment tool. It is best to make your own by grinding down some inexpensive non-metallic alignment tool available at an electronic parts store. The size is 3 mm square (slightly less than 1/8").

Vertical Size Adjustment

Vertical size is adjusted by the height control located on the front panel. This control has sufficient range to set the raster height anywhere between underscan and normal 3% overscan. The height control has no effect on the vertical linearity.

Vertical Linearity and Flat Face Adjustment

The vertical linearity of the raster is affected by two controls located on the vertical oscillator and deflection board (Board No. 162569). These are the vertical linearity control (R29), which primarily affects the top and bottom of the raster relative to the center, and the vertical flat face control (R38), which primarily affects the center of the raster relative to the top and bottom. To adjust the linearity, proceed as follows:

1. With the monitor operating, connect a crosshatch composite video signal to the input. Terminate the input in 75 ohms and set the sync selector switch for internal sync.

2. Adjust the height control, located on the front panel, for the desired raster height.

3. Adjust both the vertical linearity and vertical flat face control (R29 and R38 on the 162569 board) to obtain uniform spacing of the horizontal lines in the crosshatch pattern. This initial adjustment should be done by the visual, or "eyeball" method.

4. Using a tape measure or ruler, measure the vertical spacing between each adjacent pair of horizontal lines in the pattern (the height of each square). Write down each of these measurements.

5. Add up each of the numbers recorded in step 3 and divide the total by the number of measurements taken. This gives the average, or nominal spacing between the horizontal lines.

6. Verify that the vertical linearity is such that the height of each square is within ± 10% of nominal. If necessary, touch up the adjustment of the vertical flat face control, as necessary, to meet this requirement.

Focus and Dynamic Focus Adjustments

All SNA series monitors are electrostatically focused. Focus is adjusted by the FOCUS control located on the front panel. In addition, the larger screen versions of the SNA also have a dynamic focus circuit to improve the CRT edge focus. Focus adjustment instructions are given in the following steps. Step 1 applies to all SNA models. The remaining steps apply only to those models which have the dynamic focus circuit.

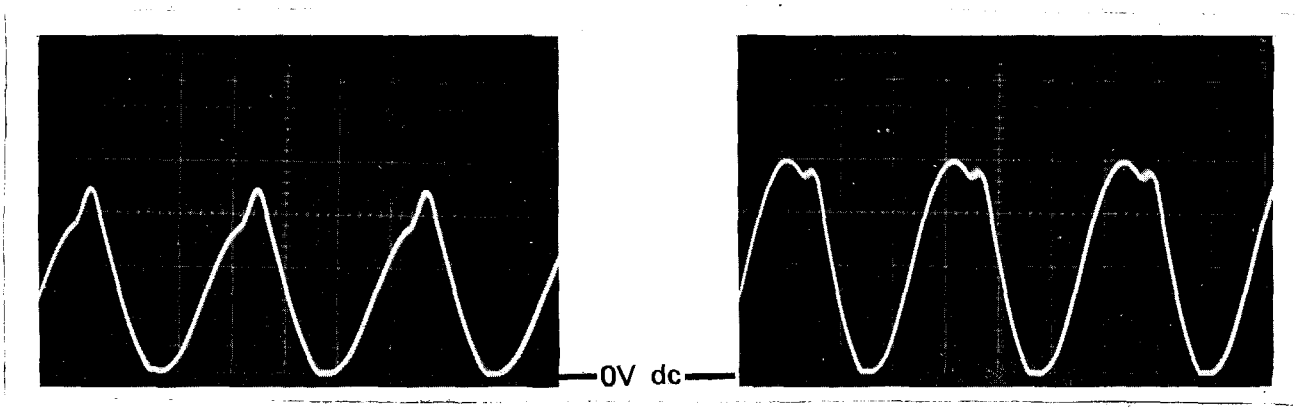
1. With a video signal applied to the monitor, as outlined for the previous adjustment procedures, adjust the brightness control for the desired brightness level. Adjust the focus control for best focus (sharpest raster lines) at the center of the CRT screen.

2. Connect the vertical input of an oscilloscope to the G3 (focus) terminal of the CRT. This connection can be made at the dynamic focus circuit board (Board No. 162511). Use a 100X low capacitance probe on the scope input.

3. Adjust the sweep timing of the oscilloscope to display one or two raster scan lines and observe the waveshape of the focus voltage. It should be essentially "U" shaped, with the two sides of equal amplitude and the tops along the same graticule line on the scope screen (See Figure 3-3.)

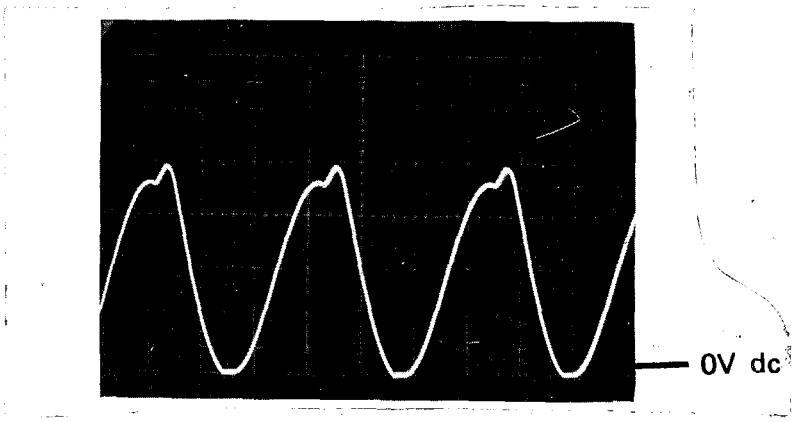
4. If necessary, adjust the dynamic focus symmetry coil (L5 on the chassis) so that the waveshape is as described in step 3.

5. Disconnect the oscilloscope from the CRT focus electrode and adjust the dynamic focus amplitude control, located on the back panel of the chassis, for optimum focus of the raster lines at the two edges of the screen along the horizontal center line (not in the corners).



a. Incorrect Symmetry Adjustment (impedance too low)

b. Incorrect Symmetry Adjustment (impedance too high)



c. Correct Symmetry Adjustment

Figure 3-3. Dynamic Focus Waveforms

Sync Gain Adjustment

A sync gain control, R36, was added to the 162438, 162644 and 162654 video and sync input boards in later production units. This control is not present in early units. The purpose of this control is to allow adjustment of the sync amplifier gain to improve the operation of the monitor on non-standard or "industrial" sync. All units are shipped from the factory with the control preset to the center. The control should only be adjusted when sync problems occur, and it should be adjusted for best sync performance from the available signal. For operation from standard EIA or CCIR sync signals, preset the control to the center.

Video Gain Adjustment

A video gain control, R26, was added to the 162398 video amplifier board in later production units. This is a contrast limiting control. It is set at the factory to optimize the monitor for a standard 1 volt p-p video input signal. It is adjusted, when necessary, to optimize the contrast performance of the monitor for signals of amplitude other than the standard 1 volt p-p. The control is adjusted, with the front panel contrast control at maximum, so that the picture displayed on the monitor screen is just at the smear point. (Smear occurs when the output signal from the video amplifier at the CRT cathode is too great, and the video output amplifier becomes overloaded.)

Horizontal Phase Adjustment

The horizontal phase control, R58 on the 162393 board, is adjusted, when necessary, to center the displayed video signal on the raster. Normally, any misadjustment of this control will have such a minor effect as to be unnoticeable, except when the monitor is used in the underscan mode.

Differential Balance Adjustment

```
**********************************
*                                *
*  THIS ADJUSTMENT APPLIES ONLY  *
*  TO UNITS EQUIPPED FOR DIFFER- *
*  ENTIAL INPUT OPERATION. THESE *
*  UNITS CONTAIN EITHER A TYPE   *
*  162438-003 (EARLY PRODUCTION) *
*  OR A TYPE 162644 (LATE PRODUC-*
*  TION) DIFFERENTIAL INPUT CIR- *
*  CUIT BOARD.                   *
*                                *
**********************************
```

1. Connect the monitor and test set-up as shown in figure 3-4. Adjust the common mode input signal (by adjusting the variable autotransformer) to 1 volt p-p.

2. Connect the oscilloscope input to the high side of the contrast control.

3. Adjust the differential balance control, R10 on the 162644 (or the 162442) board, so that the power line frequency sine wave signal seen on the oscilloscope is minimized.

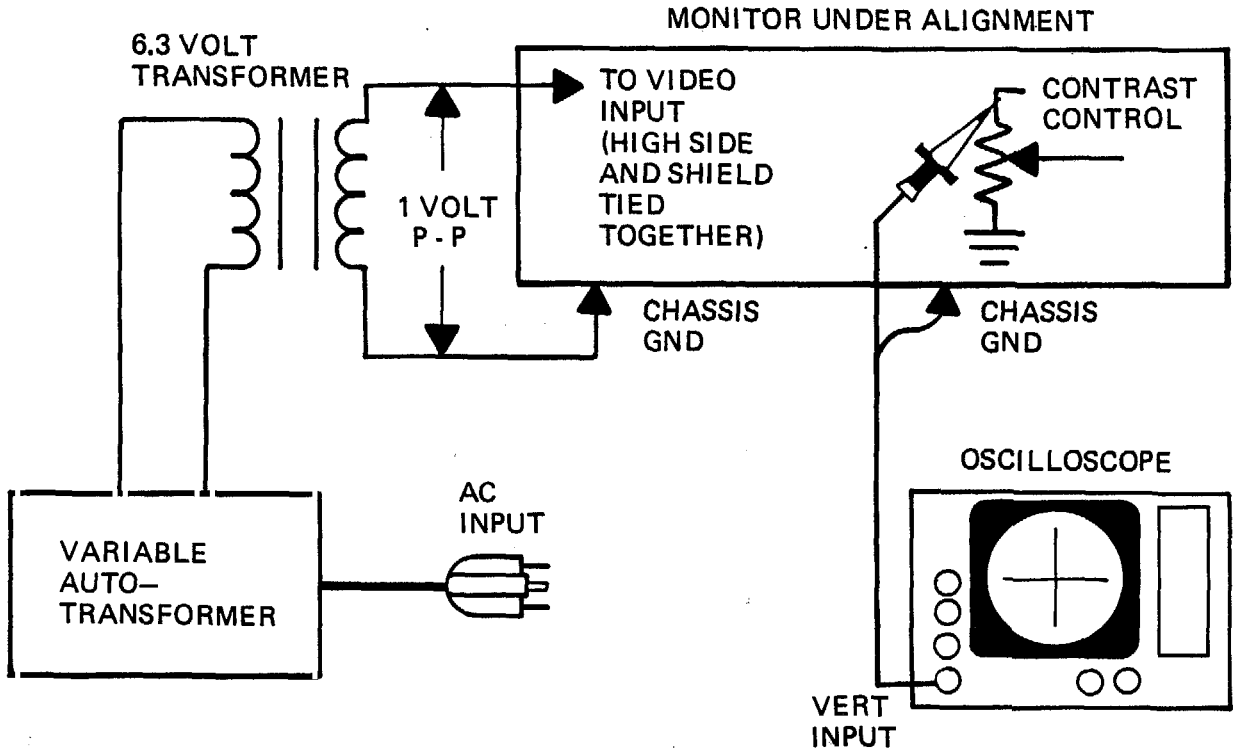


Figure 3-4. Test Setup for Differential Balance Adjustment

Alignment of the Video Amplifier Circuit

* * * * * *
* CAUTION *
* * * * * *

ALIGNMENT OF THE VIDEO AMPLIFIER BANDPASS REQUIRES A WIDE BAND (50 MHz) OSCILLOSCOPE WITH A 100X LOW CAPACITANCE PROBE, A VIDEO SWEEP GENERATOR, AND A SYNC COMPOSITOR. REFER TO FIGURE 3-5 AND TABLE 3-1. DO NOT ATTEMPT ALIGNMENT OF THE VIDEO AMPLIFIER BANDPASS UNLESS THIS EQUIPMENT IS AVAILABLE.

Standard versions of the SNA series monitors are supplied with a video channel having a 10 MHz bandpass. These units use a type 162398-001 video output amplifier board. Many special order versions of the SNA, supplied for operation at high line rates, contain a 20 MHz video output amplifier board. This is board number 162521-001. The same basic alignment alignment procedure applies to both the 10 MHz and 20 MHz units. The differences are in the passband and in the component designator numbers on the circuit board. The alignment steps that follow apply to either configuration. Where designators or frequencies differ, information contained in parenthesis ( ) applies only to the 10 MHz units containing the 162398-001 board. Information contained in brackets [ ] applies only to the 20 MHz units containing the 162521-001 board.

1. Connect the output of the video sweep generator through the sync compositor to the monitor input and terminate the input in 75 ohms. Refer to Figure 3-5.

2. Set the sync selector switch on the back of the monitor for internal sync operation.

3. Adjust the sweep generator for 1 volt p-p output and to sweep from close to dc to (20) [30] MHz. Apply markers every 5 MHz.

4. Connect the oscilloscope vertical channel to the CRT cathode through a 100X low capacitance probe.

5. Adjust the amplitude of the sweep generator output to obtain approximately 30 volts p-p average signal level at the CRT cathode. This is the amplitude of the sweep signal only, not including sync.

6. Adjust (C6) [C9 & C10] on the (162398-001) [162521-001] circuit board and L4 on the chassis to obtain the flattest possible output from dc to (10) [20] MHz. In making these adjustments, the capacitance will affect the mid-band and the inductance will affect the high end. [On the 162521-001 board, C9 affects the negative-going part of the signal and C10 affects the positive-going part.]

7. After the bandpass has been adjusted so that the envelope appears as flat as possible, adjust the vertical gain vernier on the oscilloscope so that the average of the peak to peak envelope displayed is 6 divisions high on the scope graticule. Verify that no bulge is greater than 6.7 divisions p-p and that no dip is less than 5.3 divisions p-p from dc to (10) [20] MHz. When this condition is met, the bandwidth will be within ±1 dB over the entire passband.

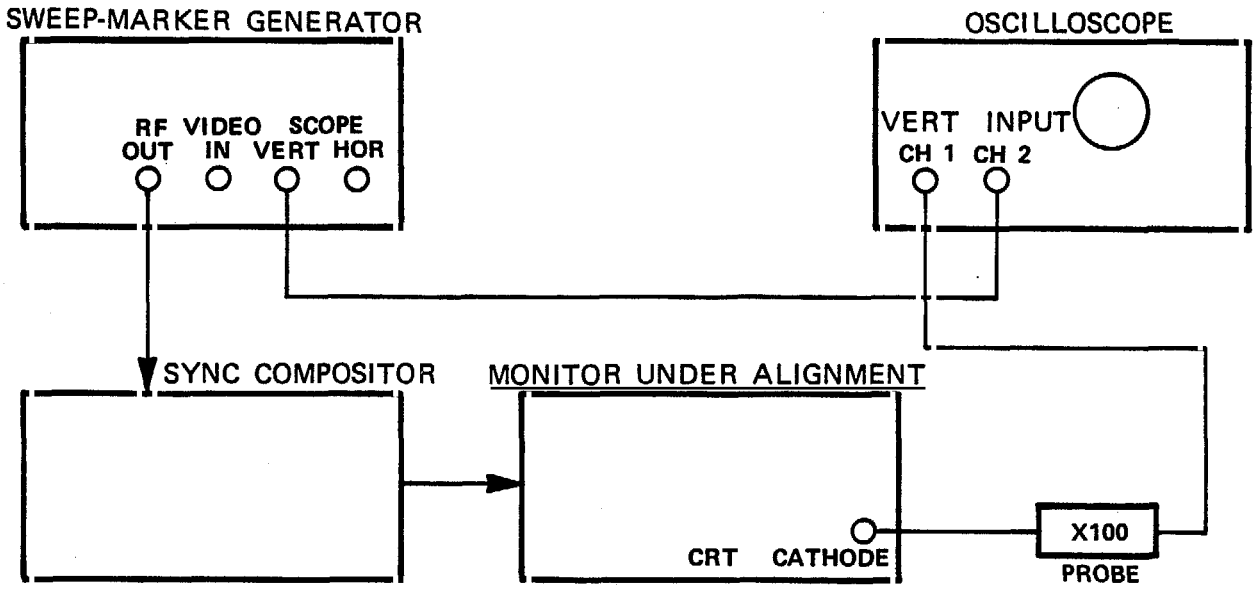


Figure 3-5. Hookup of Sweep Generator, Sync Compositor and Monitor Under Alignment

Alignment of the Video Amplifier Circuit (Continued)

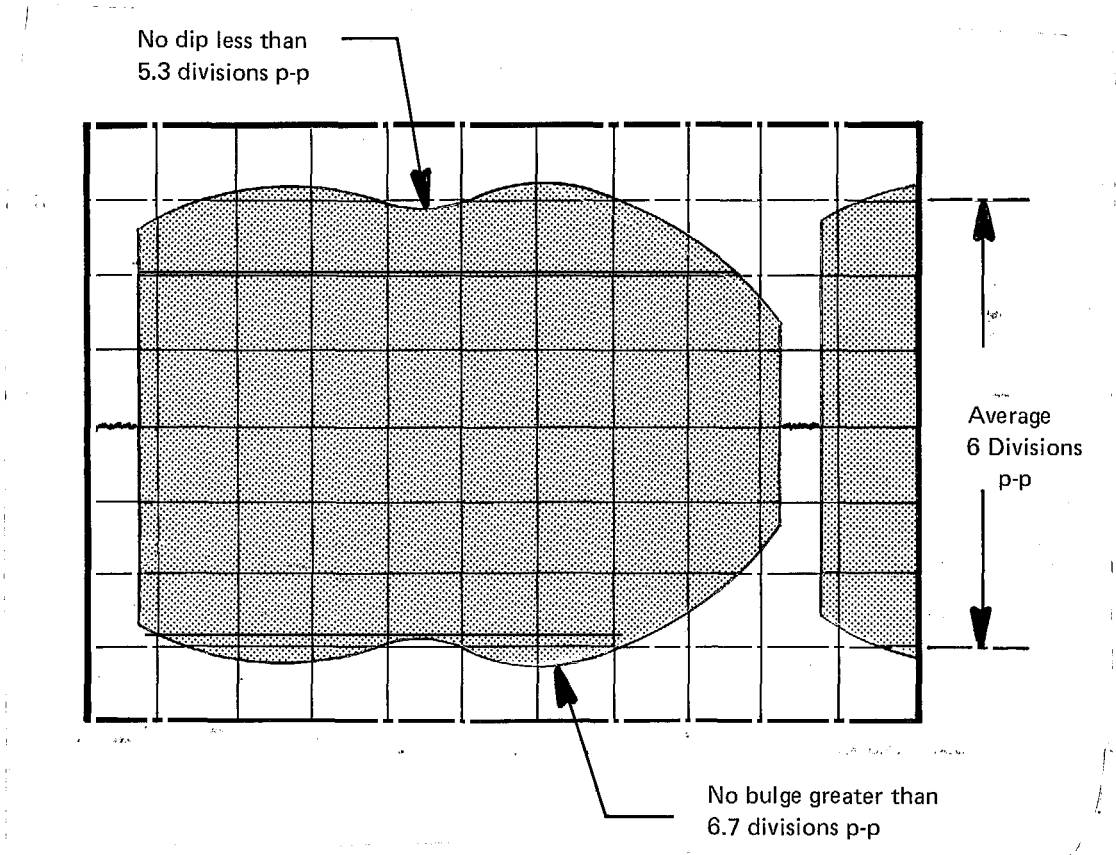


Figure 3-6. Sweep Alignment Response

CRITICAL X-RAY SAFETY INSTRUCTIONS

For All SNA Series Units

* * * * * *
* WARNING *
* * * * * *

THE FOLLOWING LISTED COMPONENTS ARE CRITICAL TO THE SAFE OPERATION OF SNA SERIES UNITS SO THEY WILL NOT EMIT HARMFUL LEVELS OF X-RADIATION. REPLACEMENT OF ANY OF THE FOLLOWING LISTED COMPONENTS MUST BE DONE USING AN EXACT EQUIVALENT COMPONENT. THE USE OF AN IMPROPER SUBSTITUTE COMPONENT MAY RESULT IN EXCESSIVE AND DANGEROUS X-RADIATION.

CRITICAL X-RAY SAFETY INSTRUCTIONS (CONTINUED)

Designation	Description	Part Number
On the Power Supply Regulator Module, Board Number 162400		
*R10	Potentiometer, B+ adjust, 2k	928490-202
*CR8	Diode, zener, 24V, 1N5359B	206019
In the Standard Line Rate High Voltage Module Number 106339		
C4,C5,C8	Capacitor, .0018mfd, 1600V, ±5%, Polypropylene	210734-001
CR2	Diode, damper, RS372	206009
Q201	Transistorm BU208, NPN	207260
T1	Transformer, driver	965071
T2	Transformer, flyback	782068
R201	Tripler bleeder	844151
In the High Line Rate High Voltage Module Number 106368		
C4,C5	Capacitor, .0015mfd, 1.5kV, ±10%, Polypropylene	210782-002
CR1	Diode, damper, RS372	206009
Q201	Transistor, BU208, NPN	207260
T1	Transformer, driver	965071
T2	Transformer, flyback	782070
R201	Tripler bleeder	844151

*Note: If any of the above components in the power supply regulator board is replaced, the B+ and +24V must be measured and, if necessary, readjusted to assure continued safety. The correct procedure is as follows:

Measurement and Adjustment of B+ and +24V

A. Equipment Required

1. Digital multimeter, 3 1/2 digit, accurate to within ±0.5% of reading or better. Fluke, model 8000A or equivalent.

2. Variable autotransformer (Variac)

B. Procedure

1. Plug the monitor into a source of power line (mains) voltage via the variable autotransformer and turn it on.

Measurement and Adjustment of B+ and +24V (Continued)

B. Procedure (Continued)

2. Adjust the variable autotransformer for a reading of 117VAC on the DMM.

3. Connect the digital multimeter (set for dc volts) between ground (chassis) and the B+ terminal on the power supply regulator board.

4. Adjust pot R10 on the 162400 board to set the voltage to +75.0 ±0.8V.

5. After the B+ has been adjusted, heat seal the adjustment control (R10) so it cannot be readjusted. This can be done by melting the plastic with the tip of a hot soldering iron.

NOTE: The +24V output of the regulator board must have a level of 24.0 ±1.2V.

SAFETY CRITICAL COMPONENTS

Designation	Description	Part Number
F1	Fuse, power (Varies with model)	
T1	Transformer, power (standard before Oct., 1977)	965070
T1	Transformer, power, tapped primary (optional before October, 1977)	965098
T1	Transformer, power, with thermal breaker	965112
V1	CRT(picture tube)	
Y1	Yoke, deflection (Part number depends on CRT size)	
	Power line cord	987013-003

NOTES

SECTION 4

CHASSIS VIEWS

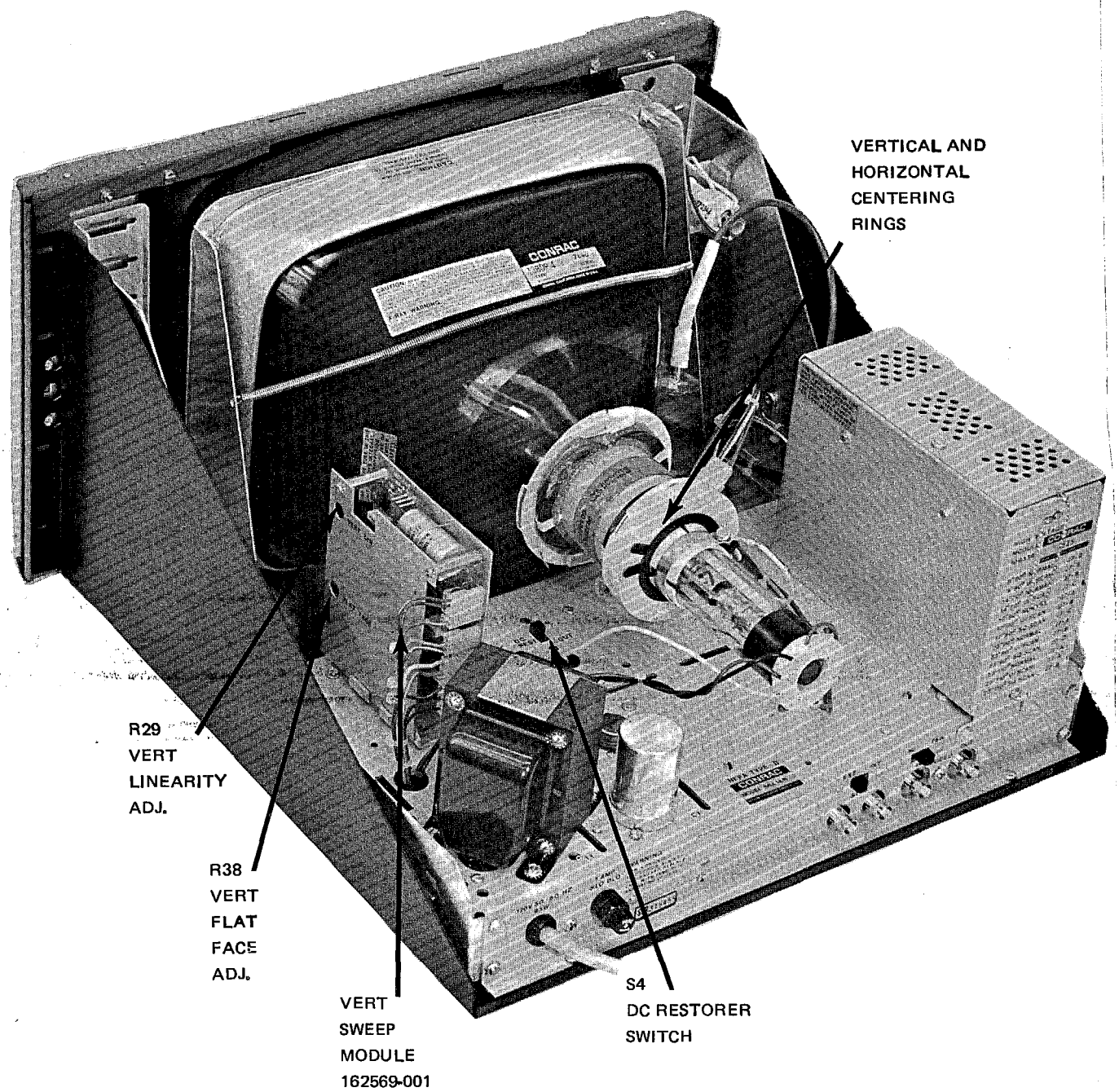


Figure 4-1. SNA-14R/15R, Right Rear View

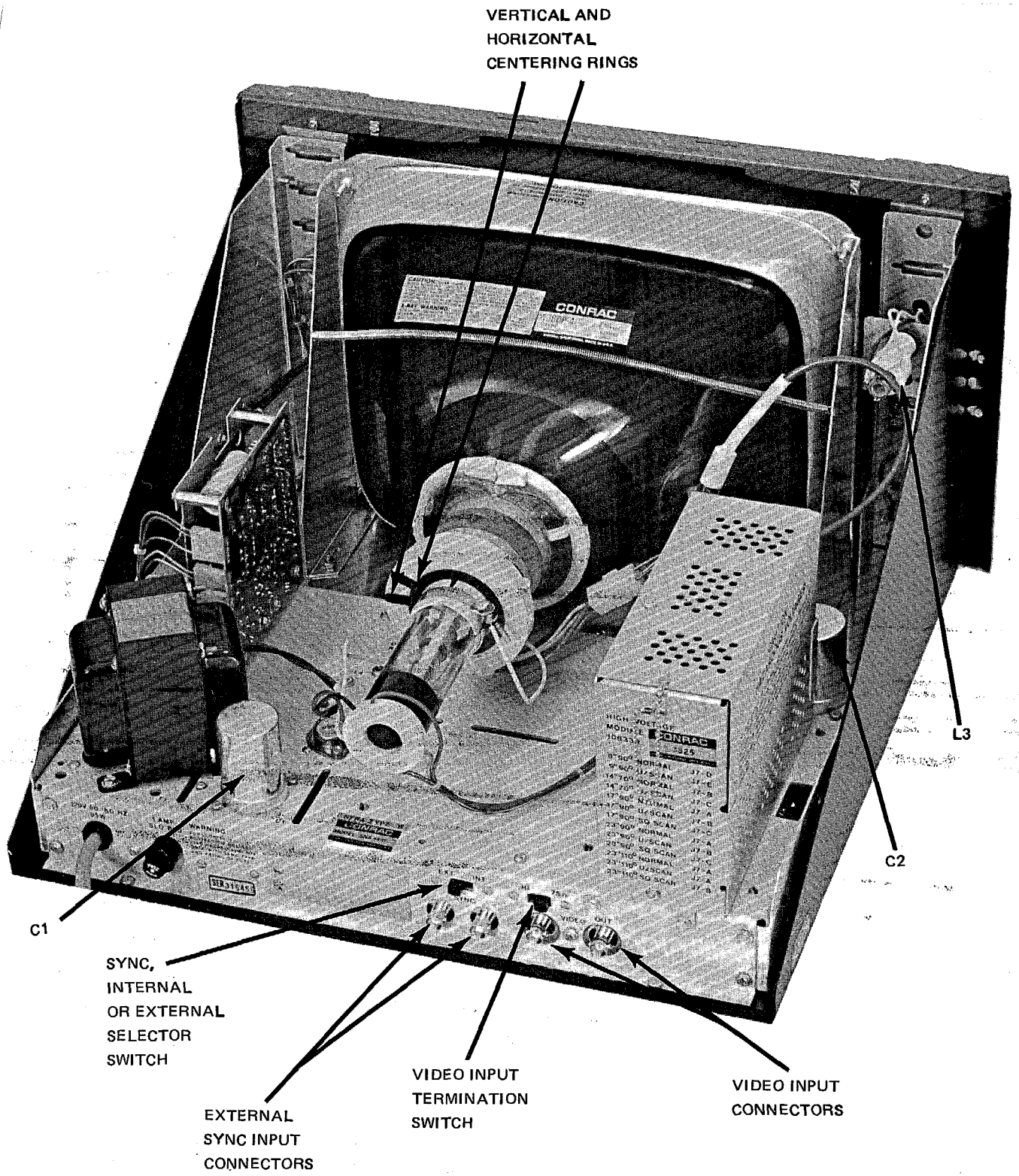


Figure 4-2. SNA-14R/15R, Left Rear View

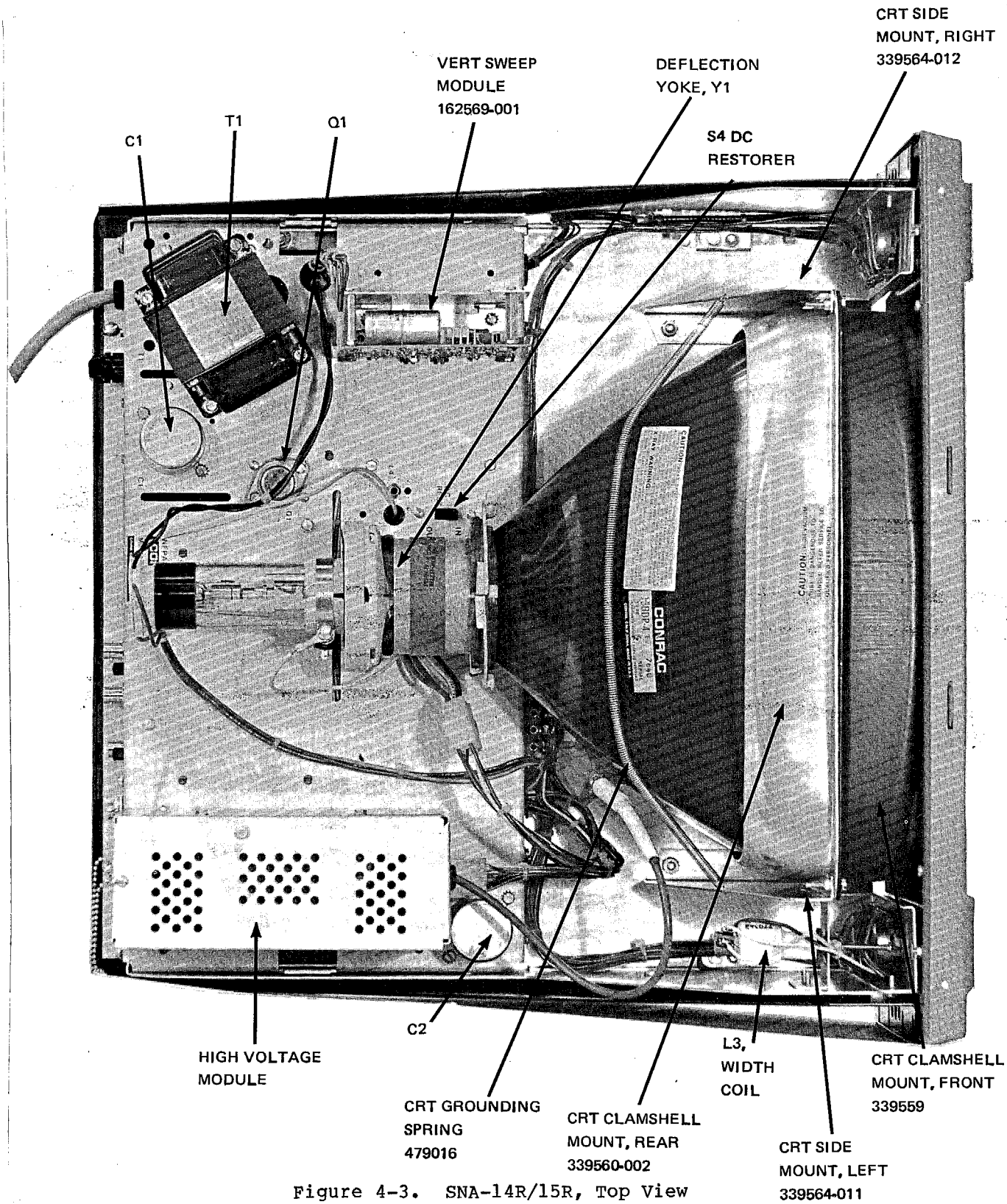


Figure 4-3. SNA-14R/15R, Top View

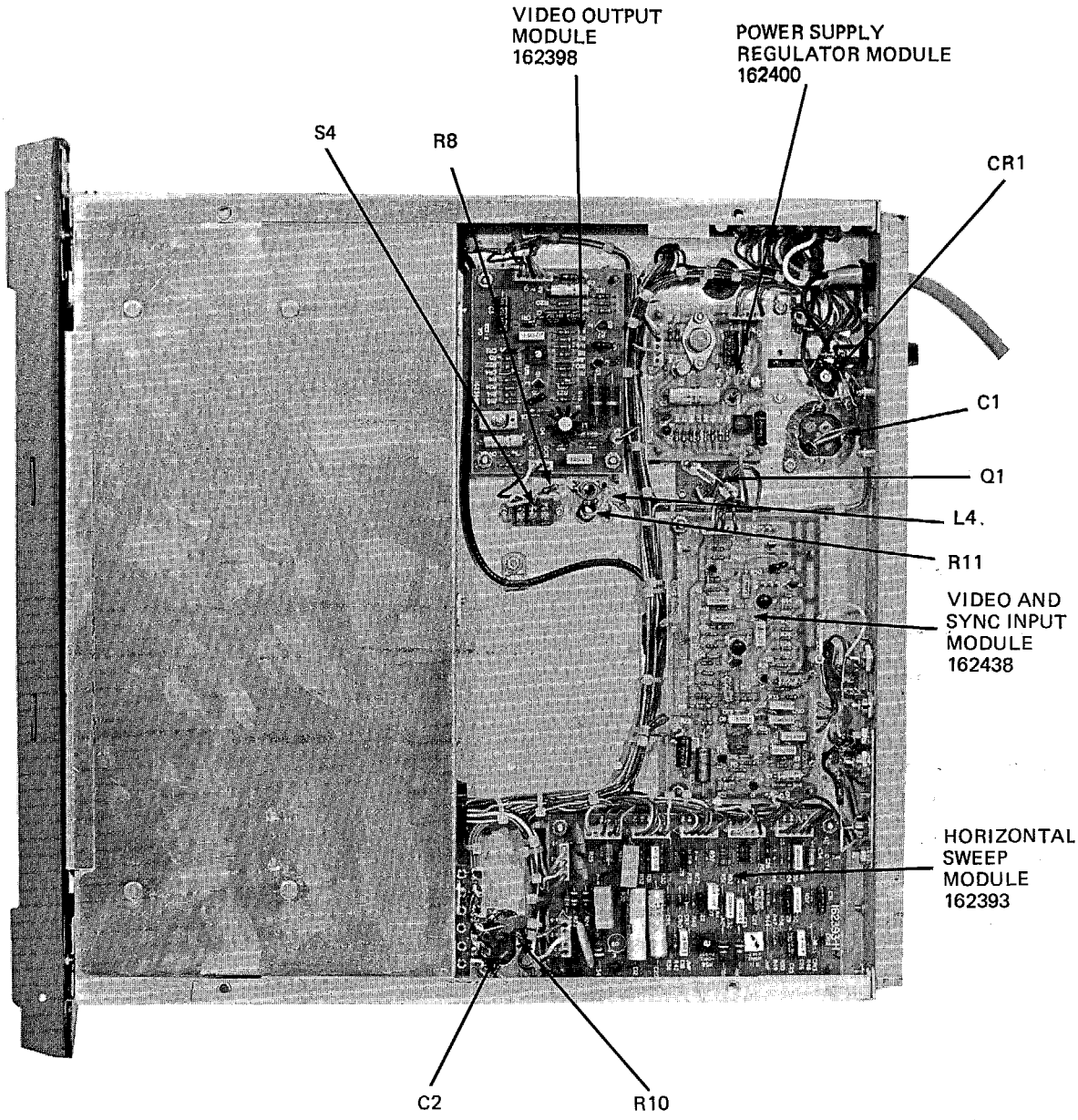


Figure 4-4. SNA-14R/15R, Chassis, Bottom View

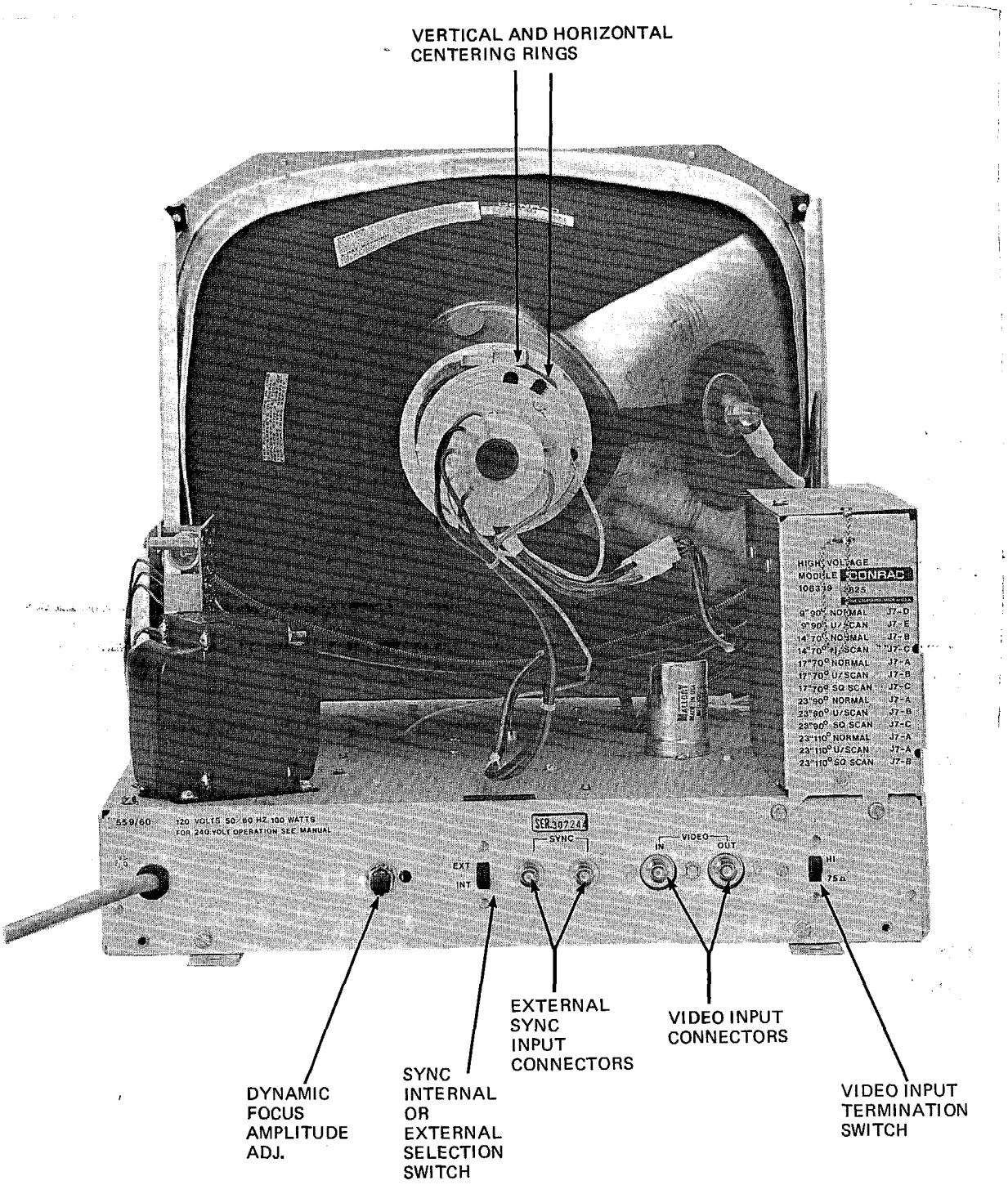


Figure 4-5. SNA-17, Chassis, Rear View (SNA-23 uses same chassis)

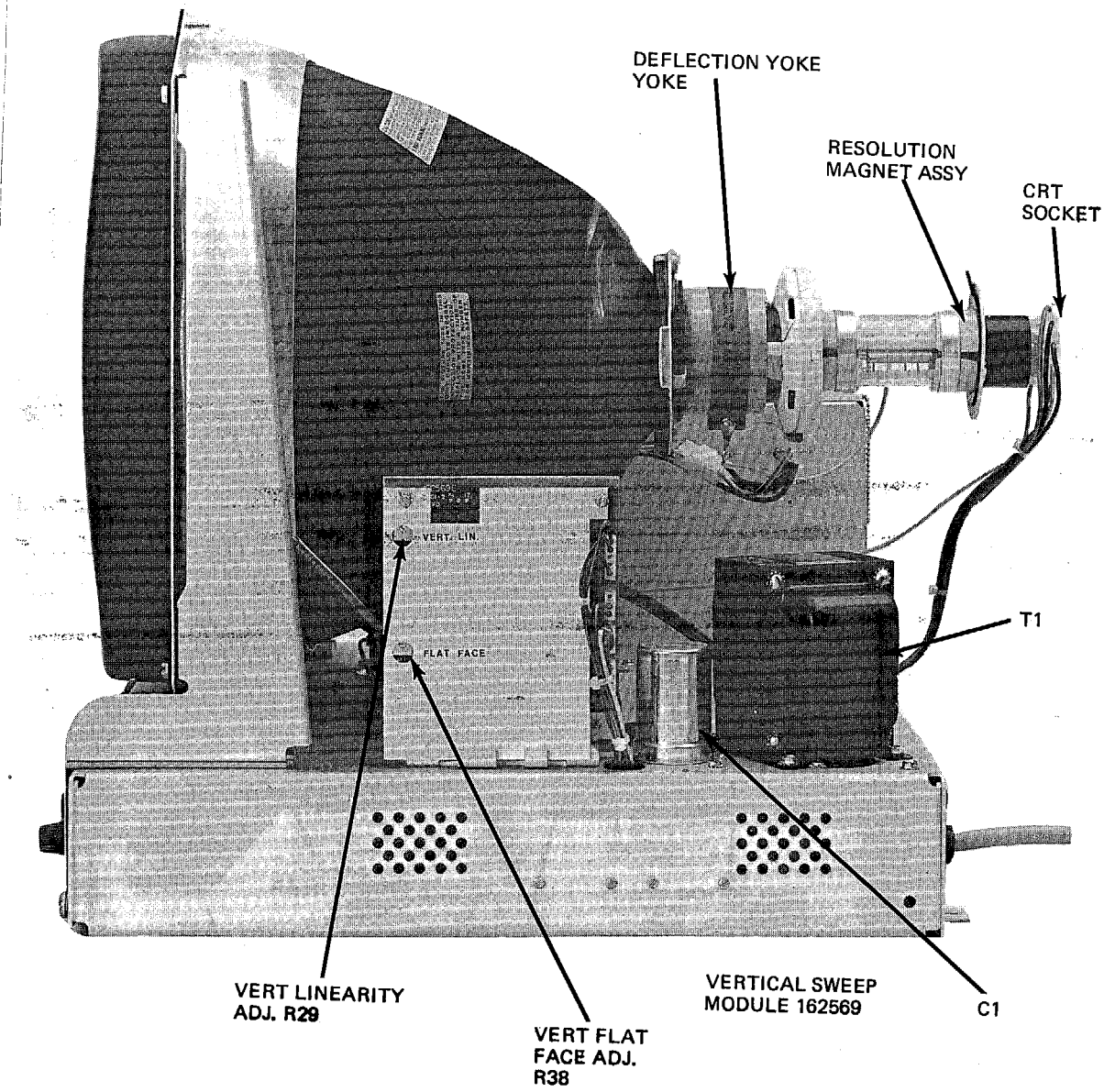


Figure 4-6. SNA-17, Chassis, Right Side View (SNA-23 uses same chassis)

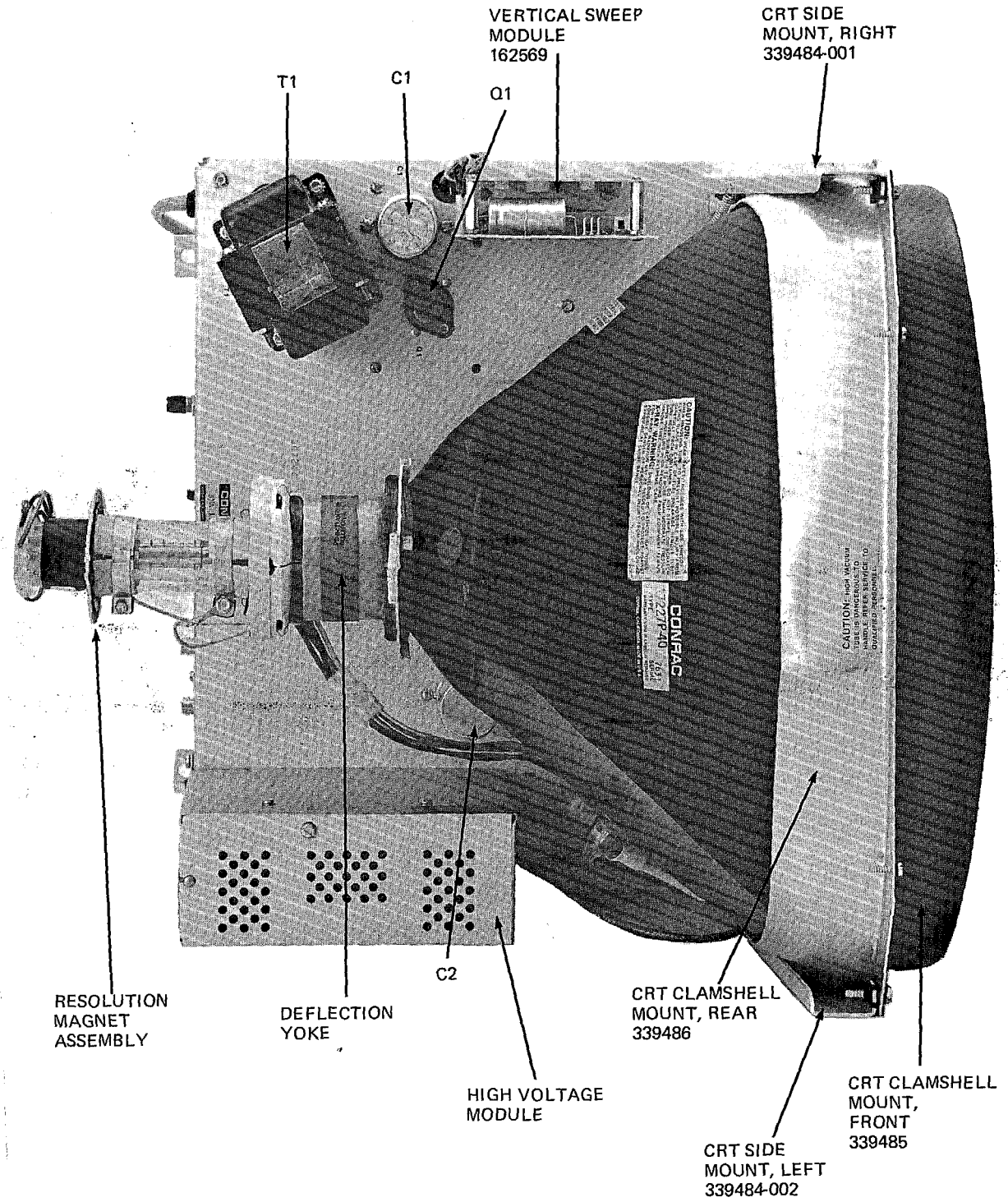


Figure 4-7. SNA-17, Chassis, Top View (SNA-23 uses same chassis)

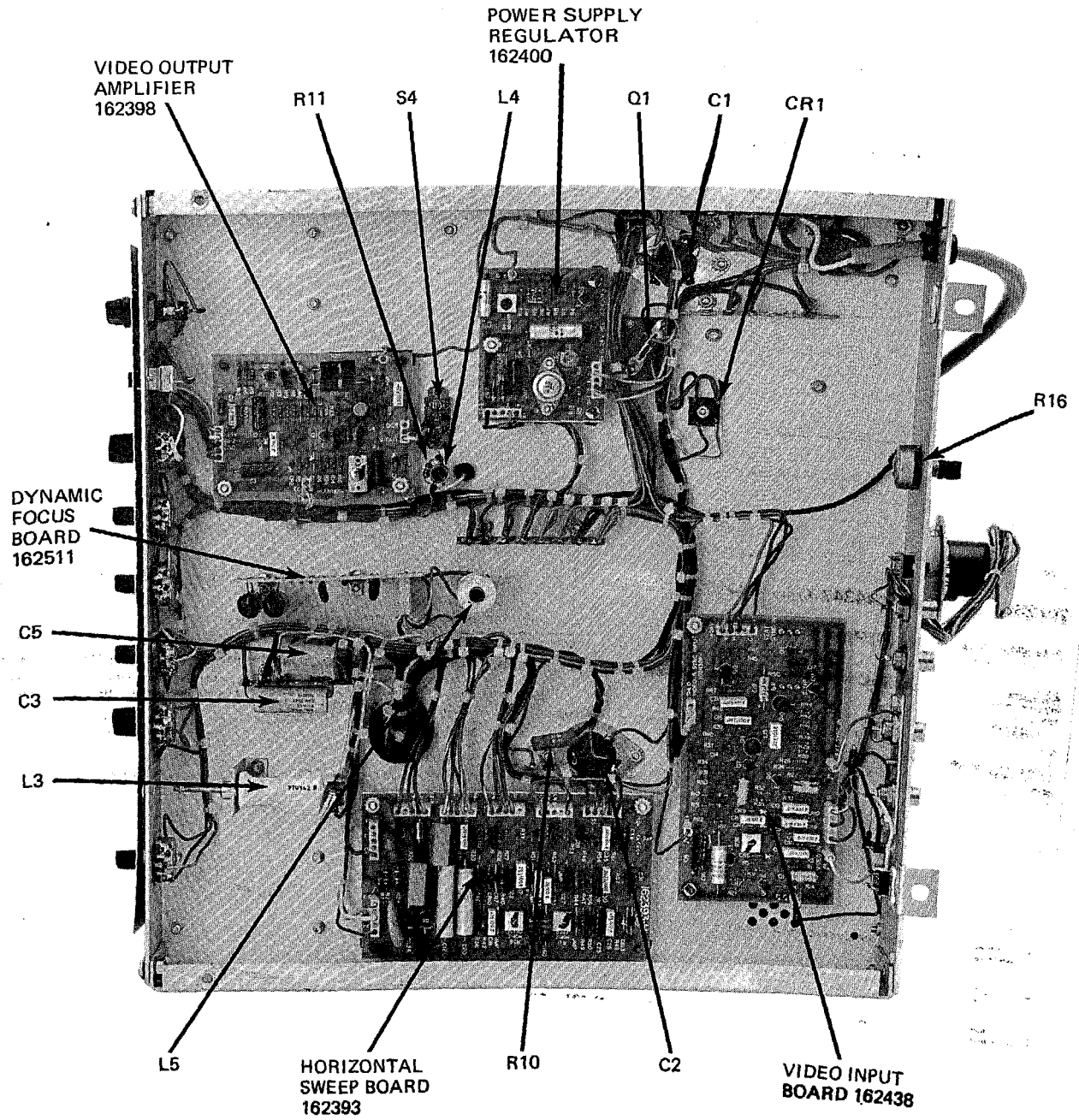


Figure 4-8. SNA-17, Chassis, Bottom View (SNA-23 uses same chassis)

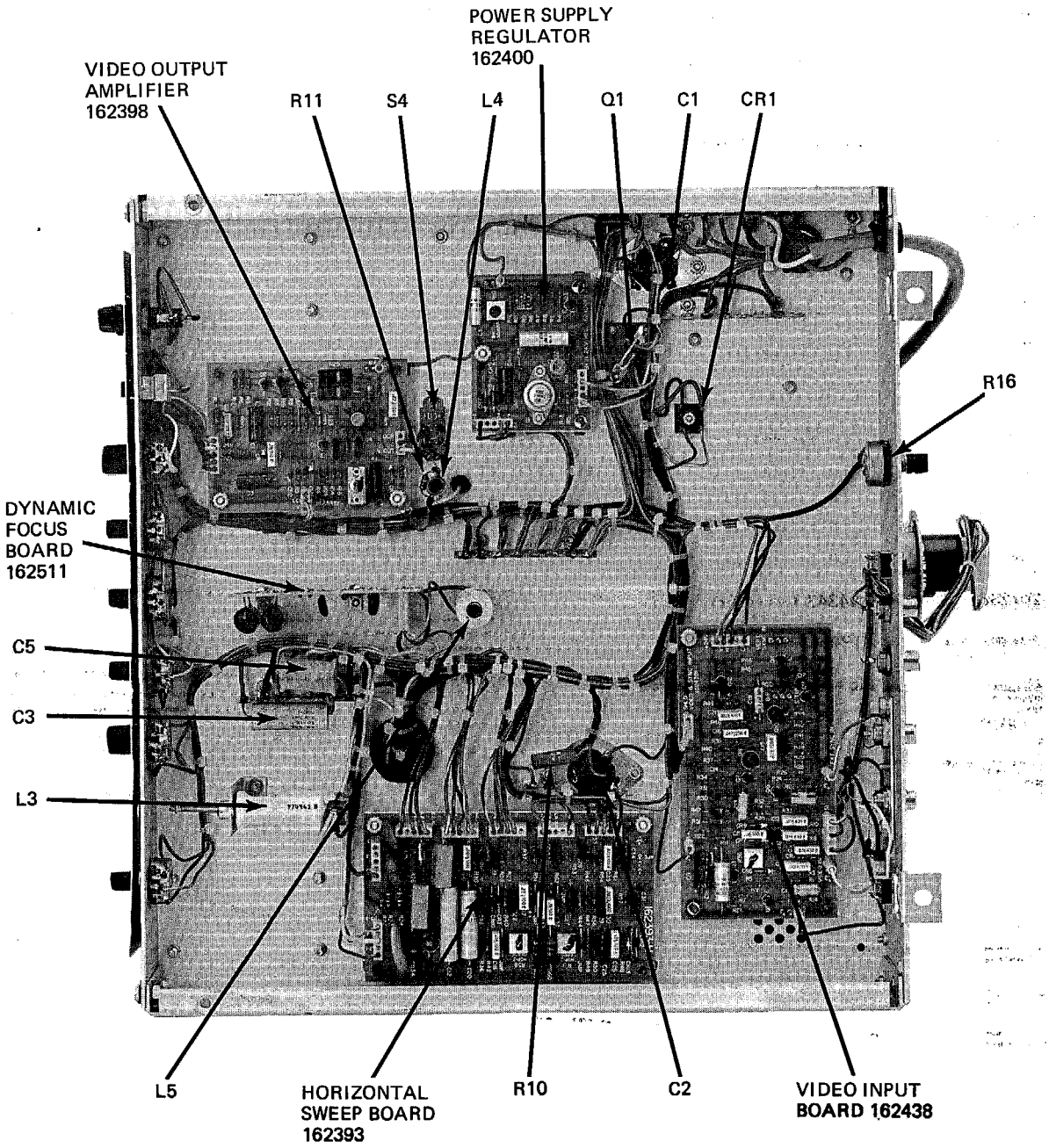


Figure 4-8. SNA-17, Chassis, Bottom View (SNA-23 uses same chassis)

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